

PROJECT REPORT

Auto Ticket Classification using Flow Designer

Team ID: SWTID-2026-6524

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1. INTRODUCTION

1.1 Project Overview

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems, and slow computers. Currently, IT staff manually reviews each request and assigns a category, which is time-consuming and inefficient.

This project automates ticket classification using ServiceNow Flow Designer by analyzing keywords in the incident's short description and description fields.

1.2 Purpose

- Automatically classify incidents at the time of creation.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Implement a no-code, easily maintainable solution.

2. IDEATION PHASE

2.1 Problem Statement

The school IT helpdesk receives multiple support requests daily from students and teachers related to common issues such as network connectivity, hardware failures, account access problems, and system performance issues.

Currently, IT staff manually review each ticket to identify the issue type, select the correct category and subcategory, and inform the caller about ticket creation. This manual process is time-consuming, error-prone, and not scalable as the number of requests increases.

2.2 Empathy Map Canvas

Empathy Map Canvas content is not provided in the project details.

2.3 Brainstorming

Brainstorming content is not provided in the project details.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Customer Journey Map content is not provided in the project details.

3.2 Solution Requirement

- Automatically classify IT tickets based on the issue description.
- Assign both Category and Subcategory without manual intervention.
- Support dependent choice logic between Category and Subcategory.
- Send an automated email notification to the caller upon ticket creation.
- Store ticket information in a structured and standardized format.
- Ensure easy maintenance and future scalability.

3.3 Data Flow Diagram

Data Flow Diagram content is not provided in the project details.

3.4 Technology Stack

The project uses ServiceNow Flow Designer and ServiceNow configuration elements including a custom table, choice fields, reference fields, Update Sets, email notification, and system records.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution fits the identified problem by automating the classification of school IT support tickets. It reduces manual classification work and improves routing efficiency.

4.2 Proposed Solution

When a student creates a new IT support ticket, they only need to enter the Caller Name and Short Description. Once the ticket is submitted, a Flow Designer automation evaluates the Short Description for predefined keywords and automatically assigns the appropriate Category and Subcategory.

- WiFi → Category: Network; Subcategory: WiFi.
- Projector → Category: Hardware; Subcategory: Projector.
- Password or Login → Category: Access; Subcategory: Forgot Password.
- Slow or Hanging → Category: Performance; Subcategory: Slow Computer.

After classification, an email notification is sent to the caller. The implementation is described as no-code, maintainable, and scalable.

4.3 Solution Architecture

The flow uses a Record Created trigger on the Incident Workflow table with the condition that Category is empty. Flow Logic checks the Short Description. Based on the matching condition, an Update Record action assigns Category and Subcategory. A Send Email action then sends a ticket confirmation to the caller.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project is organized into requirement analysis and planning, backend development and configuration, automation using Flow Designer and email notification, testing/validation/security, and deployment/final documentation.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The project includes data validation checks for mandatory fields, auto-number generation without duplication, accurate Category and Subcategory values, and correct reference-field resolution.

Test Scenario 1: Short Description: WiFi not working in library. Result: Category → Network; Subcategory → WiFi; email sent to the caller.

Test Scenario 2: Short Description: Projector not turning on. Result: Category → Hardware; Subcategory → Projector; email sent to the caller.

7. RESULTS

7.1 Output Screenshots

No images are included in this final report, as requested.

The tested flow successfully classifies the specified ticket types and sends the corresponding email notification.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces manual effort in ticket categorization.
- Improves ticket routing efficiency.
- Provides consistent classification.
- Supports automated email notification.
- Uses a no-code Flow Designer approach.
- Designed for maintainability and future scalability.

Disadvantages

The supplied project details do not specify a separate list of disadvantages.

9. CONCLUSION

The Auto Ticket Classification using Flow Designer project delivers an end-to-end automation solution for a school IT helpdesk. It automatically classifies IT tickets based on the short description, improves data consistency through dependent choice fields, and sends an email acknowledgement to the caller.

The project demonstrates how ServiceNow Flow Designer can be used to build a scalable and maintainable automation solution without complex scripting or machine learning.

10. FUTURE SCOPE

The solution can be extended to include assignment automation, SLA tracking, or predictive intelligence.

11. APPENDIX

Source Code (if any)

No separate source-code file is available.

Dataset Link

No dataset link is available.

GitHub & Project Demo Link

GitHub: <https://github.com/VidhyaSanthalingam/Auto-Ticket-Classification-using-Flow-Designer-NM>

Project Demo / Drive: [Google Drive Project Link](#)